How Much of LEV's Error Is the Catalogue's? An Empirical Evaluation of NIST's Likely Exploited Vulnerabilities Metric

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Abstract

NIST's Cybersecurity White Paper 41 proposed Likely Exploited Vulnerabilities (LEV), a probability that a CVE has already been exploited, computed by compounding a CVE's history of EPSS scores. The white paper proposed using LEV to find exploited vulnerabilities missing from CISA's Known Exploited Vulnerabilities (KEV) catalogue, and noted that measuring its performance would need outside data. We report what we believe is the first empirical evaluation of LEV. We reimplement the metric, reproduce all four average-LEV values the white paper publishes to within 11%, and test its calibration on 105,250 CVEs. Against KEV, LEV over-predicts by 17.2x. Most of that gap has causes other than LEV's core logic. EPSS was retrained during the observation window, and one retraining tripled cohort scores overnight; 26.9% of LEV's probability mass is this version artifact. LEV also grows without saturating as the analysis date advances, contrary to the white paper's premise that EPSS decays for old CVEs. To separate LEV's error from KEV's incompleteness we estimate, with capture-recapture methods over three exploitation lists, how much exploitation no list recorded. By counting alone, 30-58% of the over-prediction measured against KEV is catalogue incompleteness. Where the estimate is identified (LEV >= 0.9), LEV still over-predicts true exploitation by about 2x. A monotone correction fitted one year earlier cuts calibration error 11-17x on CVEs it never saw, but drifts, because the same LEV means different things at different CVE ages.

1 Introduction

Defenders cannot patch everything, so they rank. Two public resources dominate that ranking. The Exploit Prediction Scoring System (EPSS) gives each CVE a daily probability of exploitation in the next 30 days [2, 3]. CISA's Known Exploited Vulnerabilities catalogue (KEV) lists CVEs with confirmed exploitation [4]. EPSS looks forward and KEV looks back; neither says how likely it is that a given CVE has *already* been exploited without anyone recording it.

Mell and Spring proposed LEV to fill that gap [1]. LEV treats each 30-day EPSS score as the probability of exploitation in that window and combines the windows since the CVE first appeared:

$$\mathrm{LEV}(v, d_0, d_n) = 1 - \prod_i \left(1 - \mathrm{epss}(v, d_i) \cdot w(d_i, d_n)\right),$$

where $d_i = d_0 + 30i$ and w is 1 for full windows and the remaining fraction for the last one. The white paper suggests two uses: flagging high-LEV CVEs absent from KEV as likely exploited, and measuring how complete KEV is. It also states that assessing LEV's performance requires collaboration with parties holding exploitation data [1]. We found no published evaluation (Section 8).

The obvious evaluation compares LEV with KEV. It cannot separate two stories. If LEV says 0.99 and the CVE is not in KEV, either LEV is wrong or KEV missed it, and the second story is the one the white paper proposes LEV for. A KEV-only evaluation therefore measures LEV's error and KEV's blindness together, in unknown proportion.

We address this in two phases. Phase 1 evaluates LEV against KEV and takes the measured gap apart into causes we can observe directly in EPSS data. Phase 2 adds independent exploitation lists and estimates, with capture-recapture methods from epidemiology and human-rights statistics, the exploitation none of them recorded. That estimate is what lets LEV's error be separated from KEV's.

Our contributions:

1. **Reproduction and calibration.** We reimplement LEV, reproduce the white paper's four published average-LEV values to within 11%, and show that against KEV it over-predicts by 17.2x while still ranking CVEs into three distinguishable levels.
2. **EPSS version drift.** EPSS was retrained inside the observation window. Model version strings are build dates, not deployment dates; located by day, one retraining multiplied cohort scores 3.06x overnight. LEV compounds across such boundaries, and 26.9% of its probability mass is this artifact.
3. **No saturation.** For fixed cohorts LEV grows convexly with the analysis date and shows no sign of levelling off. Cohort EPSS rises rather than decays, contrary to the white paper's premise.
4. **Catalogue incompleteness, estimated.** After auditing candidate lists (two that appear independent of KEV contain all of it), we estimate true exploitation with log-linear capture-recapture models and a ladder of progressively stronger assumptions. Counting alone attributes 30-58% of the KEV-measured gap to catalogue incompleteness. At LEV >= 0.9, LEV still over-predicts estimated true exploitation about 2x.
5. **A validated correction.** We fit a monotone map from LEV to observed exploitation and validate it prospectively. It cuts calibration error 11-17x on unseen CVEs a year later but drifts, which exposes a further property of LEV: its meaning depends on CVE age.

All data snapshots, hashes and code are released, and one command regenerates every number in this paper (Section 10).

2 Background

2.1 EPSS

EPSS is a gradient-boosted model that scores every published CVE daily with the probability of exploitation activity in the following 30 days [2, 3]. Its ground-truth labels come from intrusion-detection and honeypot telemetry supplied by Fortinet, AlienVault OTX, the Shadowserver Foundation and GreyNoise [3]. Its 1,477 input features include public exploit code (Exploit-DB, GitHub, Metasploit), offensive tool templates (Nuclei and others), and membership of public exploitation lists, among them CISA KEV [3]. Two facts about this matter later. KEV is an *input* to EPSS, so a CVE's KEV listing can raise its later EPSS scores. And the model is retrained periodically, which can move every score in the corpus at once.

2.2 LEV

LEV is the equation in Section 1, applied from the CVE's first EPSS date d_0 to an analysis date d_n. Each 30-day window takes the EPSS score on its first day. The white paper also defines LEV2, which uses every daily score with each converted to a one-day probability. Because LEV is a product over windows it cannot decrease as d_n advances. The white paper argues the growth should stall because EPSS scores for old, unexploited CVEs decay toward zero [1].

2.3 KEV

CISA maintains KEV under Binding Operational Directive 22-01 [4]. An entry requires reliable evidence of exploitation, an assigned CVE and clear remediation guidance. `dateAdded` records when CISA catalogued the CVE, not when exploitation began. KEV is selective by design: it serves federal remediation priorities rather than aiming at complete coverage.

3 Data

EPSS. We use 67 monthly snapshots (the first of each month, plus the dates Table 5 of [1] anchors on) from 2021-04-14 to 2026-08-01: 14,456,555 scores for 367,311 CVEs. A further 171 daily files serve two checks: they locate retraining dates, and they bound the error from monthly sampling (Section 7).

CISA KEV. catalogVersion 2026.08.25, 1,676 entries. 1,665 have EPSS data; the other 11 were never scored and cannot receive a LEV value.

CVE List v5. Release `cve_2026-08-25_1700Z`, 192,054 records from 2022 onward, for CVSS, CWE and vendor.

Exploitation lists for Phase 2. The CIRCL Vulnerability-Lookup service publishes "exploited" sightings with the source of each [5]. We retrieved all 68,191 on 2026-09-28 and classified each by its source (Section 4.6). We also retrieved ENISA's EU Vulnerability Database known-exploited dump, 1,740 entries that name their source catalogue [6].

Analysis population. Unless stated otherwise, LEV is computed at d_n = 2026-08-01. The calibration cohort is the 105,250 CVEs first scored in the EPSS v3 era (from 2023-03-07), at least six months before d_n so each had time to reach KEV. We exclude CVEs that EPSS backfilled: when EPSS v2 launched on 2022-02-04 its coverage jumped from 80,800 to 168,325 CVEs, most of them old, and taking first-EPSS date as a publication proxy would count them as new. A CVE is kept only if EPSS first scored it within one calendar year of its CVE-ID year. 495 cohort CVEs are in KEV, a base rate of 0.47%.

Provenance. Every fixed file is recorded with its SHA-256. Sources served from moving branches or live APIs (Exploit-DB, Metasploit, CIRCL, EUVD) are snapshotted and committed with the code rather than re-fetched.

4 Methods

4.1 Reimplementation and reproduction

We implement LEV exactly as specified. When a window starts on a date we do not hold, we take the next available snapshot, as NIST's own implementation does. We compare the average LEV of CVEs published since each date in Table 5 of [1] with the published values, at d_n = 2025-03-01, close to the white paper's own analysis date, which it does not state. LEV grows with d_n, so the date matters: at d_n = 2026-08-01 the first row overshoots by 296%.

4.2 Calibration

We bin the cohort into ten LEV buckets and compare each bucket's mean LEV with the fraction of its CVEs in KEV, with Wilson score intervals [7] for the observed rates. Adjacent buckets are compared with Fisher's exact test under Holm correction [8]. We report the Brier score [9] and expected

calibration error (ECE) over the same ten bins [10].

4.3 Version normalization

To remove score shifts caused by EPSS retraining, we map every score produced after the 2025-03-17 retraining back onto the preceding model's scale. Each score keeps its percentile within its own day's file, and we substitute the score at that percentile in the last pre-retraining file (2025-03-16). This removes corpus-wide shifts. It cannot separate them from genuine per-CVE improvement in the newer model, so normalized LEV is a lower bound on a correction, not a corrected value.

4.4 Analysis-date sweep

For two fixed cohorts (13,685 CVEs first scored March-September 2023, and 7,558 first scored May-September 2021) we compute LEV at every month from 2023-10-01 to 2026-08-01 and fit six growth families: linear, logarithmic, saturating, quadratic, exponential and power law. We compare them by AIC over the full sweep and within the longest stretch with no retraining.

4.5 Circularity

Because KEV is an EPSS input, we test how much listing moves scores, with two designs that cannot select on the outcome. First, for each KEV CVE we compare EPSS in the last snapshot before `dateAdded` with the first snapshot on or after it, net of non-KEV CVEs in the same starting-score decile over the same pair of snapshots. Pairs that straddle a retraining are excluded. Second, we compute each KEV member's LEV using only scores dated before its listing, which the listing cannot have influenced.

4.6 Capture-recapture

Lists. An exploitation list is a set of CVEs a source records as exploited in the wild. We consider three:

- CISA: CISA KEV.
- SENSOR: CIRCL sightings sourced from Shadowserver's honeypot feed (969 CVEs), which detects attempted exploitation directly.
- SOCIAL: CIRCL sightings from Telegram and similar channels (2,396 CVEs).

We audited every candidate for overlap with CISA KEV before use. CIRCL's aggregated KEV catalogue contains all 1,676 CISA KEV CVEs, and VulnCheck's KEV includes CISA KEV by its own documentation [11]; either used as a "second list" makes the overlap equal to CISA KEV by construction. We exclude the first and, lacking an API token, could not rebuild the second from its per-report sources. EU KEV lists 66 CVEs; 55 are also on CISA KEV, and 88% of those carry CISA's own date, so we treat it as derivative. Exploit-DB records exploit availability rather than exploitation and is excluded as a list.

Population. SENSOR begins on 2024-10-24. We restrict to the 54,612 CVEs first scored between 2024-10-24 and 2026-01-30, so every list could observe every CVE from its first score, and apply the six-month maturity rule. 570 of these appear on at least one list.

Models. Two-list estimates disagree by 2.4x (530 to 1,256), so the lists are not independent and dependence has to be modelled. We fit Poisson log-linear models to the seven observable capture patterns under every hierarchical structure of two-way interactions, assume no three-way interaction

(the standard identifying assumption [12, 13]), select by AIC, and bootstrap a 95% interval from the fitted cell means. Chao's estimator [14], a lower bound that stays valid under heterogeneous capture probabilities, is reported alongside.

When not to report an estimate. Log-linear estimates of the unseen cell can be driven entirely by the choice of interaction terms. We declare a population not identified, and report no point estimate, if models within 2 AIC of the best disagree by more than 2x [15], if the estimate exceeds the population, or if it falls below the lower end of Chao's interval.

The truth ladder. Rather than one estimate, we report LEV's over-confidence (mean LEV over the exploitation rate) against successively stronger notions of truth, each adding one assumption: KEV alone; KEV or SENSOR; the union of all three lists; the two-list Chapman estimate; Chao's bound; the log-linear estimate. Every rung is a lower bound on, or an estimate of, the number exploited, so every over-confidence figure is an upper bound on LEV's error. A reader can stop at whichever assumption they accept.

4.7 Correction curve

We fit a monotone map from LEV to observed exploitation by isotonic regression [16] and by Platt scaling [17], and a Platt variant that also takes log CVE age. There are two targets: KEV on the calibration cohort, and the union of the three lists on the Phase 2 population. All scores are on held-out CVEs, under three schemes. Five-fold cross-validation is the in-distribution case. A split by first-score date trains on older CVEs and tests on newer ones. The deciding scheme is prospective: the map is fitted at d_n = 2025-08-01 with outcomes known then, and applied at 2026-08-01 to the 44,316 CVEs first scored after the training cohort, none of which it saw.

5 Results

5.1 The reimplementation reproduces the white paper

CVEs published since	CSWP 41	ours	difference
2023-03-07	0.015	0.0162	+8%
2022-02-04	0.047	0.0479	+2%
2021-04-14	0.087	0.0838	-4%
all CVEs	0.146	0.1299	-11%

Table 1: Average LEV at d_n = 2025-03-01 against Table 5 of [1].

All four values fall within 11%. The fourth row matches only with the backfill correction of Section 3; without it we get 0.255 (+75%). This validates the implementation. It does not validate the metric, and the agreement is specific to the analysis date.

5.2 Against KEV, LEV over-predicts 17.2x but ranks

Across the 105,250-CVE cohort, mean LEV is 0.081 against a KEV rate of 0.0047: LEV over-predicts by 17.2x (Brier 0.0373, ECE 0.0763). The observed rate climbs from 0.03% in the lowest LEV bucket to 12.9% in the highest, where LEV averages 0.987 (Figure 1). Only two of the nine adjacent-bucket differences survive Holm correction, those at either end. LEV separates three levels (low, middle, high) rather than ten.

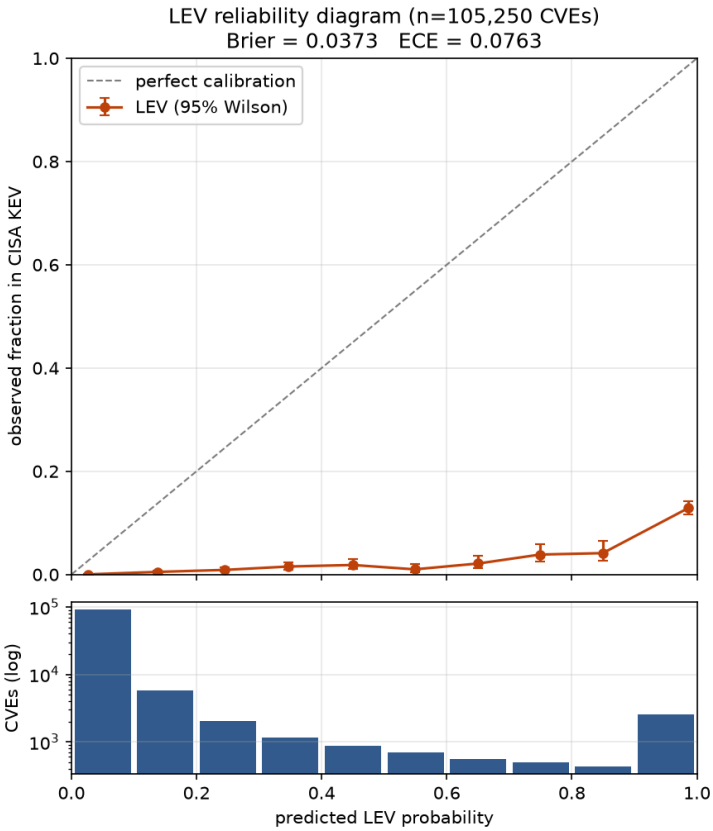


Figure 1. Reliability of LEV against CISA KEV. Error bars are 95% Wilson intervals.

5.3 EPSS retraining puts an artifact into LEV

EPSS's model version strings are build dates. Version v2025.03.14 first appears in the file for 2025-03-17. On an unchanged set of 13,552 CVEs, cohort mean EPSS goes from 0.00862 on 2025-03-16 to 0.02640 the next day, a 3.06x step (Figure 2). Anyone bracketing by the version string measures almost no change. Of the rise in cohort EPSS over 2023-2026, 64% (v3 cohort) and 70% (v1/v2 cohort) happens at retraining boundaries rather than between them.

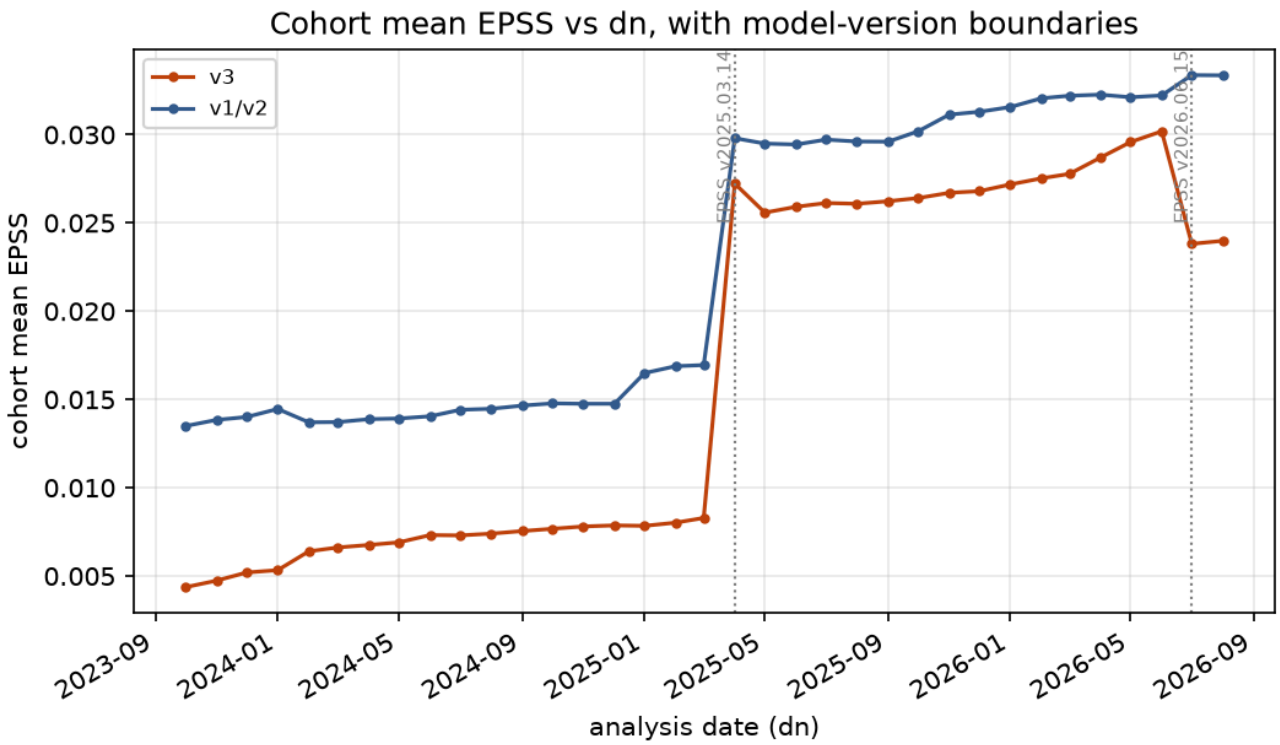


Figure 2. Cohort mean EPSS, with model-version boundaries.

LEV multiplies scores across these boundaries as if they belonged to one measurement series. When post-retraining scores are mapped back to the v3 scale (Section 4.3), mean LEV drops by 26.9%, and over-prediction against KEV falls from 17.2x to 12.6x (Brier 0.0288, ECE 0.0546). Version mixing accounts for part of the gap but leaves most of it. Normalization also moves three of the four Table 5 values toward the published figures at d_n = 2026-08-01 (mean absolute deviation 102% to 63%), without ever seeing them; the fourth, which spans pre-v3 models, moves away.

5.4 LEV grows without saturating

For the 2023 cohort, mean LEV rises from 0.008 to 0.135 between October 2023 and August 2026 (Figure 3). A quadratic fits better than linear, logarithmic or saturating growth, both over the full sweep and within the retraining-free stretch (AIC margin 12.8 over linear there), and the preference survives version normalization. Monthly increments grow rather than shrink. The premise behind the expected slowdown fails as well: cohort mean EPSS rises from 0.0044 to 0.0240 rather than decaying. If this continues, the share of the cohort above LEV 0.9 roughly doubles by 2028, and LEV loses discrimination as more of the corpus approaches 1.

Monotone growth is a property of the equation, not a finding. The empirical content is the shape: no saturation within 35 months, and no evidence of the decay the white paper relies on.

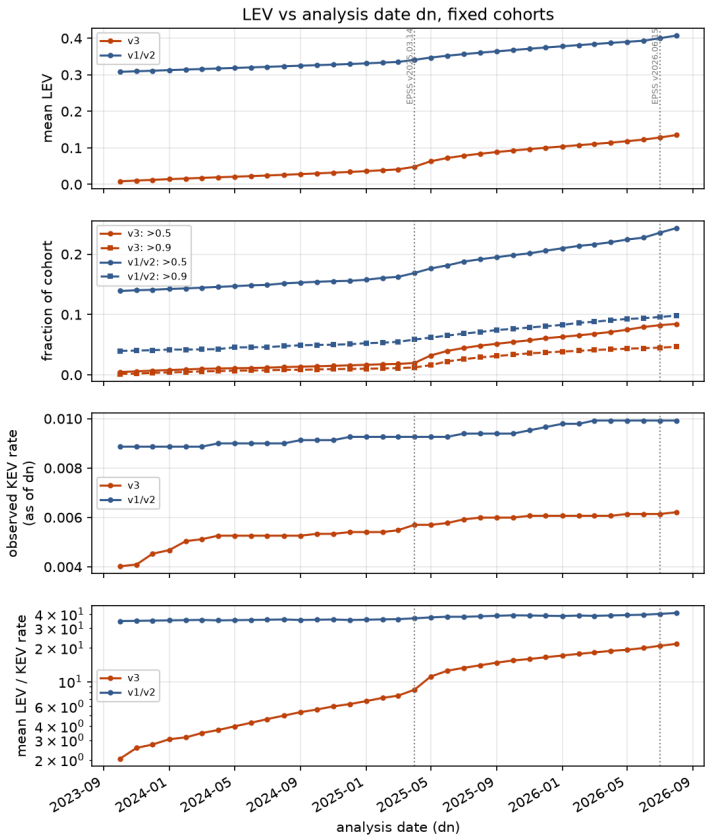


Figure 3. LEV for fixed cohorts as the analysis date advances. Dotted lines mark retrainings.

5.5 High LEV mostly follows KEV listing

At the listing date, the median KEV CVE's EPSS does not move relative to matched non-KEV controls, but a quarter of listed CVEs rise at least 2.85x more than their controls. The KEV input feature matters for a sizeable minority of listings rather than all of them.

The second test bears on the white paper's proposed use. 56% of cohort KEV members were listed in the same month EPSS first scored them, so LEV could not have flagged them first. Of the 220 with a pre-listing history (a median of 2.8 months), 15% had LEV >= 0.5 before CISA listed them, and the median was 0.005; after listing the median is 1.0. Part of that gap is elapsed time, since pre-listing LEV covers fewer windows. Either way, among the CVEs KEV did list, high LEV mostly arrived after the listing. This tells against using LEV as a leading indicator of KEV listing, and it means the high-LEV bucket's KEV concentration in Section 5.2 is partly circular.

5.6 How much of the gap is the catalogue's

Table 2 gives the truth ladder on the 54,612-CVE Phase 2 population, where KEV alone gives 7.91x.

truth	assumption added	exploited	over-confidence
CISA KEV	KEV is complete	236	7.91x
CISA or SENSOR	honeypot sightings are real	336	5.56x
union of three lists	every sighting is real	570	3.28x
Chapman (CISA, SENSOR)	these two lists independent	730	2.56x
Chao lower bound	closed population	887	2.11x [1.89, 2.31]
log-linear, AIC-selected	no three-way interaction	1,896	0.98x [0.59, 1.38]

Table 2: LEV over-confidence against successively stronger notions of truth. Intervals are 95%.

If only KEV and honeypot sightings are trusted, over-prediction falls from 7.91x to 5.56x. If all three lists are trusted, it falls to 3.28x. So by counting alone, with no model, 30-58% of the gap measured against KEV is catalogue incompleteness; under Chao's bound it is at least 73%. Equivalently, KEV recorded at most 41% of the exploited CVEs that any list observed in this population, and at most 27% under Chao's bound.

We report but do not claim the bottom rung. It pools strata that the stratified analysis rejects as heterogeneous, and the lowest stratum is dominated by sightings we cannot vouch for. Table 3 and Figure 4 stratify by LEV.

LEV	N	union	Chao	log-linear
[0.0, 0.1)	52,019	2.93x	0.66x	not identified: below Chao's bound
[0.1, 0.9)	1,957	4.59x	3.41x	not identified: models span 143 to 2,997
[0.9, 1.0]	636	2.80x	2.42x	2.07x [1.75, 2.46]

Table 3: Over-confidence by LEV stratum.

Only the top stratum supports an estimate. There LEV averages 0.985, and we estimate 47% [40%, 56%] of those CVEs were exploited. Every rung above counting agrees on 2.1-2.4x. The middle stratum's best models tie on AIC while implying anywhere from 143 to 2,997 exploited CVEs, more than the stratum holds, and three lists cannot choose between them. In the lowest stratum 86% of the CVEs seen by exactly one list were seen only on Telegram, where actors also post mass lists of CVEs they claim to use; the model estimate falls below Chao's bound there, and we do not report it.

Two checks. Shadowserver is one of EPSS's label sources, so EPSS is trained to predict roughly what SENSOR records. Without SENSOR, over-confidence is 3.85x against the union and 2.71x against Chao's bound: LEV looks somewhat worse without the list it was trained toward, as expected, and the conclusion holds. Second, the listing boost of Section 5.5 lifts some KEV members into the top stratum. Those CVEs are true positives, so without the boost the top stratum's exploited share would likely be lower, and the 2x figure is probably conservative.

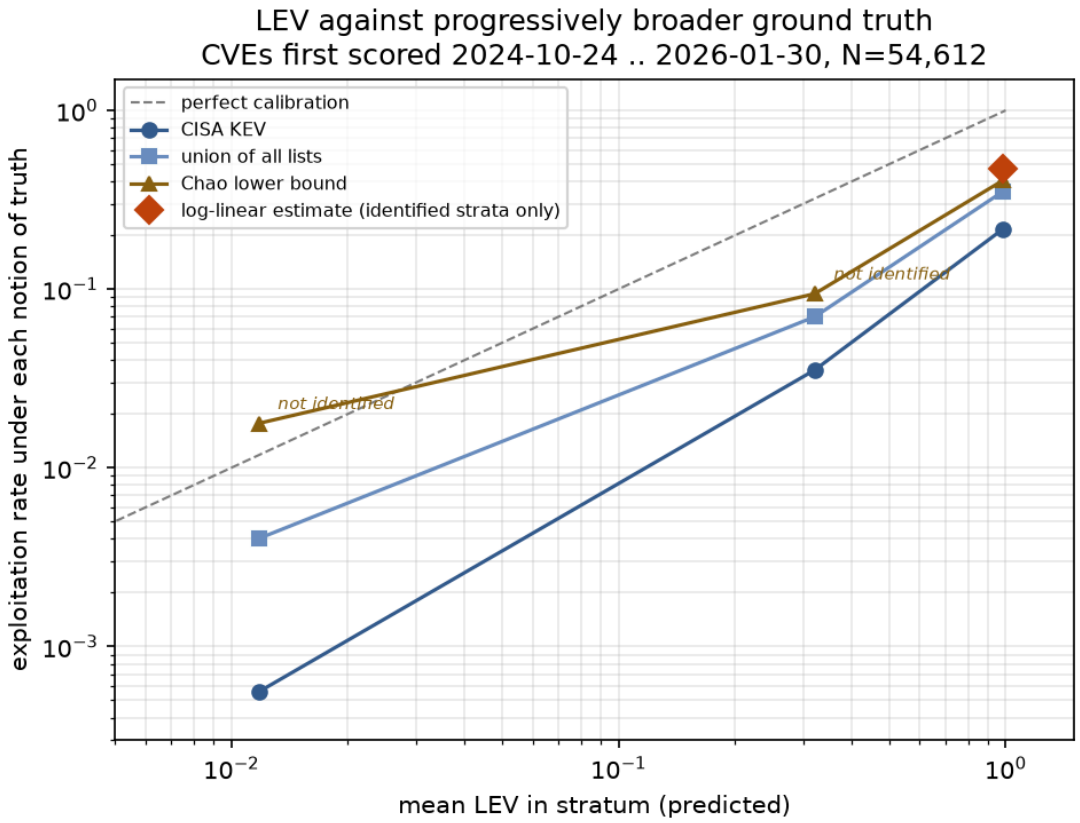


Figure 4. LEV against progressively broader ground truth, by LEV stratum.

5.7 A correction curve, and why it drifts

method	Brier	ECE	predicted / observed
raw LEV	0.0112	0.0229	6.28x
isotonic	0.0036	0.0020	0.61x
Platt	0.0037	0.0015	0.66x
Platt with CVE age	0.0036	0.0014	0.72x

Table 4: Prospective validation against KEV. Fitted at 2025-08-01; scored at 2026-08-01 on 44,316 CVEs the fit never saw.

Under cross-validation any of the three maps restores calibration (predicted over observed 1.00-1.01x). Prospectively, each cuts calibration error 11-17x, but all under-predict by 28-39% a year later. Against the union target, isotonic regression transfers almost exactly (0.95x; ECE 0.0163 to 0.0021; Figure 5).

The drift has a cause we can name. An old CVE reaches a given LEV through many windows of modest EPSS, and a new one through a few high scores. The second is stronger evidence, so the same LEV value carries a different probability at different ages. A Platt model that also takes log age recovers about a quarter of the drift. This is the non-saturation of Section 5.4 seen from another side: LEV's scale depends on how long it has been accumulating.

LEV	corrected to KEV	corrected to union of lists
0.10	0.8%	4.9%
0.50	2.7%	9.2%
0.90	6.8%	12.8%
0.99	8.2%	24.2%
1.00	27.8%	66.7%

Table 5: The isotonic correction, fitted on all data. Both targets are catalogue appearance and so are lower bounds on true exploitation.

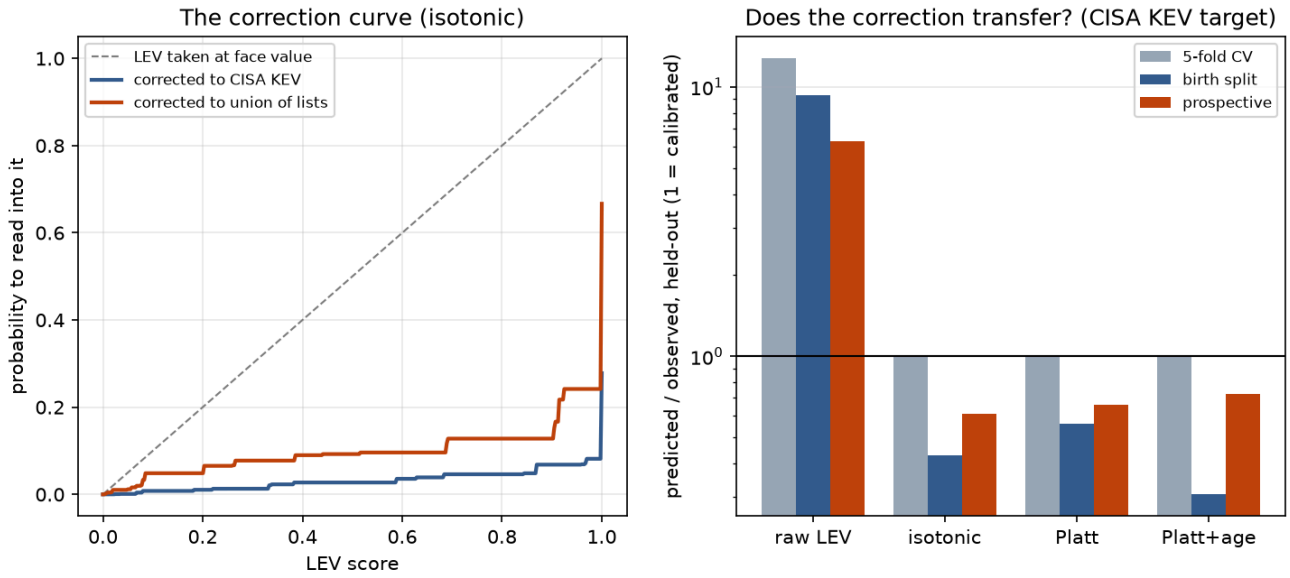


Figure 5. Left: the correction curves. Right: held-out predicted/observed under each validation scheme.

6 Discussion

What LEV is good for. LEV ranks. Its extremes separate: the top bucket's KEV rate is nearly 500 times the bottom bucket's. Its values should not be read as probabilities. A LEV of 0.9 corresponds to about 7% chance of KEV listing and about 13% chance of appearing on any of our three lists.

What LEV is not good for yet. The white paper proposes LEV for finding exploited CVEs that KEV misses. Our results cut both ways on that. LEV's apparent over-confidence is substantially KEV's incompleteness, which is the premise of the proposal. But among CVEs KEV did list, high LEV mostly came after the listing, so LEV did not anticipate them.

What would make LEV more reliable. Four changes follow from the results. Compute LEV within a single EPSS model version, or normalize across versions, since retraining otherwise enters as evidence. State the analysis date with every published LEV value, since values are not comparable across dates. Stop accumulating, or decay, windows for CVEs with no recent evidence, or accept that LEV drifts toward 1. And validate against more than KEV: KEV is both an EPSS input and incomplete.

KEV coverage. Empirical Security reports that KEV covers at most about 8.7% of CVEs with observed exploitation activity, from proprietary telemetry [18]. Our bounds are upper bounds on coverage, from open data, on a younger population: at most 41% by counting, at most 27% under

Chao's bound, and 12% [7%, 18%] from the pooled model we do not claim. The directions agree. The levels differ as they should, since our lists are fewer and narrower than one vendor's full telemetry.

7 Threats to validity

We record 51 threats in the released repository. The ones that bear most on the conclusions:

- **Monthly sampling.** LEV's windows are 30 days and our snapshots are about 30.4 days apart. On 479 CVEs stratified by LEV decile, LEV from true daily windows differs from ours by 3.9% of mean LEV (population-weighted), with no consistent sign. The check covers the calibration cohort, not the older CVEs behind the Table 5 rows.
- **Publication-date proxy.** We use first EPSS appearance as publication date, with a backfill filter. Without the filter, Table 5's "all CVEs" row misses by 75%.
- **Circularity.** KEV is an EPSS input (Section 5.5), and Shadowserver is an EPSS label source. We measure the first and drop the second as a check; we cannot remove either from EPSS.
- **List dependence.** The log-linear model assumes no three-way interaction, which three lists cannot test. We report estimates only where identified.
- **False sightings.** Telegram sightings include claims we cannot verify. Every headline figure is also given with that list excluded.
- **A short, young population.** SENSOR starts in October 2024, so Phase 2 covers CVEs first scored over 15 months, each observed for 6-21 months.
- **Normalization.** Percentile mapping removes corpus-wide shift and cannot separate it from genuine improvement in the newer model.
- **KEV dates.** `dateAdded` is a cataloguing date. Timing comparisons are lagged by an unknown amount.
- **Search scope.** Our search for prior work was targeted, not systematic.

8 Related work

EPSS and its evaluation. EPSS was introduced by Jacobs et al. [2] and extended with a larger feature set in version 3 [3]. Parla evaluates EPSS on high-severity CVEs in KEV [19]. These works evaluate the forward-looking score; none evaluates LEV.

LEV. Beyond the white paper [1], an independent reimplementation by the Risk Based Prioritization project questions LEV's small-probability approximation and its independence assumption, and disputes the white paper's reading of EPSS for previously exploited CVEs [20]. It does not validate LEV against exploitation data. We found no empirical evaluation of LEV in arXiv or the security literature up to September 2026.

Prioritization with KEV as ground truth. Recent frameworks combine CVSS, EPSS and KEV [21] or build composite risk indicators evaluated on KEV [22]. ThreatLens trains a ranking model on KEV entries as weak supervision [23]. Our results suggest caution with any of these: KEV is incomplete in a vendor-dependent way, and it is already an EPSS input. Shaffer and Voicu model exploitation hazard with a decay calibrated on KEV timing [24]; our cohorts show EPSS rising, not decaying, over 35 months.

Multiple-systems estimation. Capture-recapture over administrative lists is standard in epidemiology [12] and has been used to estimate casualties and modern slavery, where its sensitivity to list dependence and sparse cells is well documented [25]. We found no prior application to vulnerability exploitation.

9 Conclusion

Evaluated against KEV, LEV looks badly over-confident. Much of that is not LEV's doing. EPSS retraining puts an artifact into the product LEV computes, LEV accumulates without the saturation its design assumes, and KEV records a minority of the exploitation that other lists observe. Where exploitation can be estimated, at the top of LEV's scale, LEV still over-predicts about twofold. A monotone correction brings its values much closer to observed rates, but has to be refitted as CVEs age.

A fourth independent list would identify the middle of LEV's scale, where three lists cannot. VulnCheck's KEV, rebuilt without CISA-sourced evidence, is the natural candidate; the released code includes it once an access token is supplied.

10 Data and code availability

[Repository URL]. `make fetch` downloads the fixed inputs and `make reproduce` regenerates every table and figure. SHA-256 hashes, release tags and retrieval dates for all inputs are listed in `DATA_SNAPSHOT.md`.

References

See `references.md` for full entries and verification status.

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